21/12/24

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[22EE52]

FIFTH SEMESTER B. Tech EXAMINATION, DECEMBER - 2024
CAPITAL SYSTEM ARCHITECT – SIEMENS 3

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A	**2 X 5=10**		
1.	**a.**	Function Designs are implementation independent, if?	L2	CO1	**02**
	b.	Why Metrics are calculated ?	L1	CO1	**02**
	c.	What are carrier types?	L1	CO1	**02**
	d.	At which levels can styling be applied?	L2	CO1	**02**
	e.	What happens when a functional design is associated with the platform?	L3	CO1	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	What can function allocation rules be based on in system design?	L2	CO2	**05**
		OR			
	b.	How does Platform Synthesis handle updates during system design?	L2	CO2	**05**
3.	**a.**	What key activities are involved in Platform Level Engineering?	L3	CO2	**05**
		OR			
	b.	What metrics are used in Capital Systems Architect to measure system performance?	L2	CO2	**05**
4.	**a.**	What are some example roles for users of Capital Systems Architect?	L2	CO2	**05**
		OR			
	b.	What can platforms be created for in Capital Systems Architect?	L2	CO2	**05**
5.	**a.**	What does Platform Level Engineering involve?	L2	C03	**05**
		OR			
	b.	What component types are included in the system?	L2	CO3	**05**
6.	**a.**	What does the Extensibility framework allow data to be extracted from platforms for use in?	L3	CO3	**05**
		OR			
	b.	What does logical synthesis from platforms produce?	L3	CO3	**05**

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7.	a.	List the objects connected by the pathways	L3	CO4	05
		OR			
	b.	What are the possible synthesize options for a carrier?	L2	CO4	05
		PART-C	**6 X 10=60**		
8.	a.	What are the possible ways in which signals connecting functions can be handled in a system?	L2	CO5	10
		OR			
	b.	How does the interface map in a component manage messages and signals, and what options do you have for mapping them?	L3	CO5	10
9.	a.	Write the main features of Capital Systems Architect in terms of function and signal management?	L2	C04	10
		OR			
	b.	Name the trans command used for?	L2	C03	10
10.	a.	What are the ways signals can be mapped to carriers?	L3	C04	10
		OR			
	b.	List out the interface maps used for in a system? Interface maps are used for two primary purposes:	L3	CO5	10
11.	a.	What are constraints used for in a system design process?	L2	C02	10
		OR			
	b.	On what elements can constraints be added in a system design?	L3	C03	10
12.	a.	What can metrics measure in a system design?	L4	C02	10
		OR			
	b.	When creating a new metric, what are the elements that need to be defined?	L4	CO3	10
13.	a.	What are the DRC (Design Rule Check) modes available?	L2	CO3	10
		OR			
	b.	What do hyperlinks in DRCs point to?	L2	CO4	10